

• By defn, a free variable corresponds to a column in the matrix A which has no pivots.

• By defn, a pivot position of a matrix corresponds to a leading 1 of one of its rows in reduced row echelon form.

$$\begin{aligned} & \begin{bmatrix} 2 & -5 & 8 \\ -2 & -7 & 1 \\ 4 & 2 & 7 \end{bmatrix} \xrightarrow[\text{Row 1}]{\text{Row 2} = \text{Row 2} + \text{Row 1}} \begin{bmatrix} 2 & -5 & 8 \\ -4 & -2 & -7 \\ 4 & 2 & 7 \end{bmatrix} \xrightarrow[\text{+ Row 2}]{\text{Row 3} = \text{Row 3} - \text{Row 1}} \begin{bmatrix} 2 & -5 & 8 \\ -4 & -2 & -7 \\ 0 & 0 & 0 \end{bmatrix} \\ & \xrightarrow[\text{Row 1}]{\text{Row 2} = \text{Row 2} + 2 \times \text{Row 1}} \begin{bmatrix} 2 & -5 & 8 \\ 0 & -12 & 9 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\times (-1/12)]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 2 & -5 & 8 \\ 0 & 1 & -3/4 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{+ } 5 \times \text{Row 2}]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 2 & 0 & 17/4 \\ 0 & 1 & -3/4 \\ 0 & 0 & 0 \end{bmatrix} \\ & \xrightarrow[\times 1/2]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & 0 & 17/8 \\ 0 & 1 & -3/4 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Verification of whether the final matrix is in reduced row echelon form

1. Row 1 and 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all non-zero rows must be above the zero rows is True.

2. Row 1 Leading Element = 1 in Column 1

Row 2 Leading Element = 1 in Column 2

$\therefore$ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0

Row 3 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry, all positions below the leading entry must contain 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry, all positions above the leading entry must contain 0 is True.